

Southwest and AirTran: what the fare comparison actually measures

Case study memo | BTS ticket sample, 2007 Q1 to 2015 Q4 | Saathvik Maheshuni | September 2026

Summary

When two airlines merge, the first question is whether fares went up on the routes where they used to compete against each other. I found **24 nonstop airport pairs** where Southwest and AirTran each sold a meaningful number of tickets in the second quarter of 2009, well before the merger was announced.

The usual way to answer the question is to compare the average fare on those routes with the average fare on similar routes that only AirTran flew. For 2015, measured against the four quarters before the September 2010 announcement, that comparison gives **+1.3% (-4.9%, +8.1%)**. **That number should not be reported as a fare effect.** A route's average fare is an average across whichever airlines are selling tickets on it, and by 2015 one of those airlines is gone.

To see how much of the number that explains, I split it in two. First I froze every airline's own fare at its level before the merger and let only the passenger shares change. That alone gives **+2.0%**. What is left, the part that comes from airlines actually changing their own fares, is **-0.7%**. The mix accounts for 156% of the headline. Southwest's fares on these routes were higher than AirTran's in 2009, so when the AirTran name disappears the route average rises on its own, without anyone changing a price.

So I changed the measure. Instead of the route average I used **Southwest's own fare**, and instead of AirTran-only routes I compared against Southwest routes that AirTran never flew. No change in airline mix can move that number. Across 23 matched pairs, Southwest's own fare on the old overlap routes in 2015 was +12.0% (+6.7%, +17.9%) higher than on the comparison routes. This version also passes the checks the first one fails. Its trend before the merger is +1.4% (-1.4%, +4.5%) a year, which is flat because the range crosses zero, against +2.9% (+0.2%, +5.4%) for the route average. A fake test on an early period, where nothing should show up, gives -1.0% (-6.7%, +5.5%).

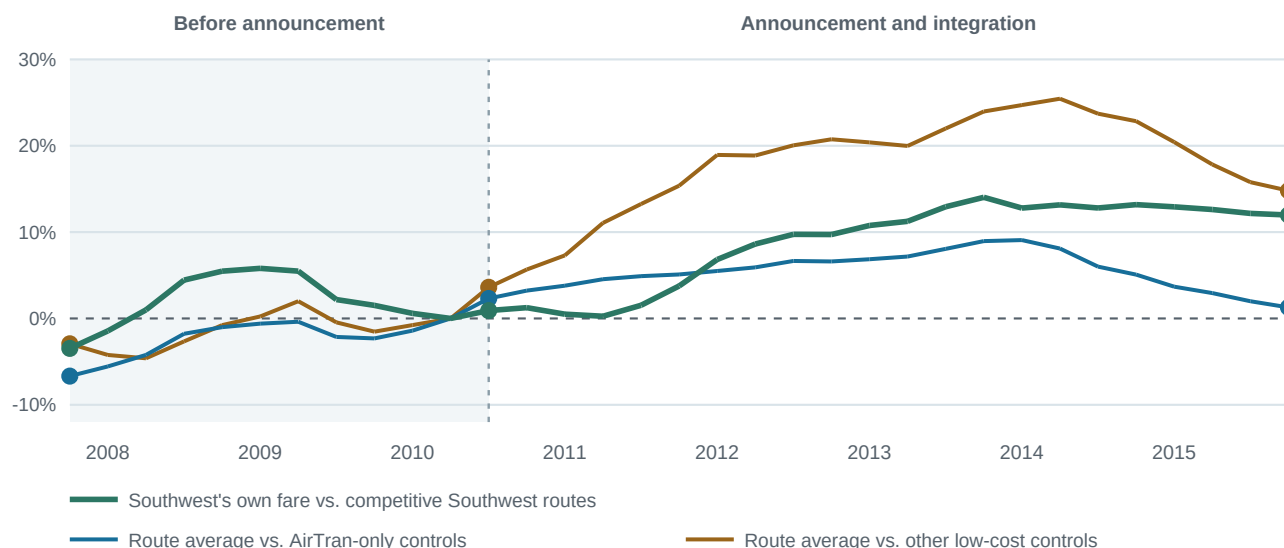
The direction looks solid. The size does not. 20 of the 23 overlap routes touch a Florida airport, but only 10 of the 137 possible comparison routes do, because Southwest's other competitive routes are mostly in the West. So the comparison ends up matching Florida routes against Western ones, and the two regions did not move together. Looking only at comparison routes, where AirTran never flew at all, Southwest's fare rose +4.5% on Florida routes and +20.8% everywhere else. That gap pushes my number *down*, not up. Forcing each route to be compared within its own region raises it to +21.2%, but only by reusing the same 8 comparison routes over and over. The honest way to report this is a positive difference of roughly 12% to 21%, not a single number.

Background

The Department of Justice closed its investigation on April 26, 2011. It said the two airlines overlapped on few nonstop routes, that it expected new service to follow, and that there were no gate or slot problems at the airports involved. [5] The merger closed on May 2, 2011, and the last scheduled AirTran flight left Atlanta on December 28, 2014. [3,4] This memo only looks at fares on routes the two airlines already shared. It says nothing about whether passengers gained from the routes Southwest added later, which was the other half of what the DOJ weighed.

Why the route average is misleading

Fare gap over time, four quarters at a time



Each point averages four quarters in a row and subtracts the gap in the four quarters before the announcement, so zero means no change from that starting point. The three lines disagree. The AirTran-only route average ends up back near where it started. The other low-cost route average sits far above it the whole time, on a comparison that is poorly matched. Southwest's own fare rises and stays up. The shaded part is before the September 27, 2010 announcement. [2]

Splitting the 2015 number into fares and mix

For the mix-only version, every airline keeps the fare it charged in the four quarters before the announcement, and only the share of passengers each one carries is allowed to change. Whatever that version shows is the work the changing mix did by itself. Whatever is left over is what airlines did to their own fares.

Comparison group	Total	From changing mix	From actual fares
AirTran-only routes	+1.3% (-5.0%, +8.0%)	+2.0% (-3.3%, +7.7%)	-0.7% (-9.0%, +7.7%)
Other low-cost routes	+14.8% (+7.5%, +22.2%)	+4.9% (+2.1%, +8.1%)	+9.4% (+3.1%, +15.5%)
AirTran-only, matched on early trend	-0.4% (-6.8%, +6.4%)	+2.3% (-3.3%, +8.2%)	-2.7% (-11.0%, +6.1%)

The parts are averages of log differences across matched pairs, so they do not add up neatly in percentage terms. Each total reproduces the published figure exactly before it is reported here. Note that the other low-cost comparison is mostly real fare change, so the three versions disagree about the cause as well as the size.

How I define an overlap route matters more than any single route

A route counts as an overlap if each airline had at least 50 sampled passengers in the second quarter of 2009. The data is a 10% sample of tickets, so 50 sampled passengers is roughly 500 real ones in a quarter. On a busy airport pair that can be a few percent of traffic. On four of the 24 routes the smaller airline carries under 10% of the passengers, as little as 3.8% on Indianapolis to Las Vegas. Asking the smaller airline to be a real presence moves the route-average number by several points. That is a much wider swing than dropping one route at a time, which only moves it between -0.3% and +2.8%.

Smaller airline's share of route traffic, 2009 Q2	Pairs	Route average	Southwest's own fare
No minimum (as built)	24	+1.3% (-4.9%, +8.1%)	+12.0% (+6.7%, +17.7%)
At least 10%	20	+3.9% (-2.6%, +11.3%)	+12.0% (+6.7%, +17.7%)
At least 20%	13	+9.6% (+1.6%, +18.6%)	+12.0% (+4.9%, +20.2%)

The two columns use different pools of comparison routes, which is why the route counts differ. The route-average number is not stable under the very definition that produced it. Southwest's own fare barely moves, and dropping one route at a time only shifts it between +10.8% and +13.0%. Neither version holds up against the regional makeup of its comparison routes, which is the real problem here and is covered below.

Data and method

Ticket records for all 36 quarters

The data is the BTS DB1BMarket file, a 10% sample of airline tickets that records passengers, the fare assigned to each trip, and the operating airline. [1] I processed every quarter from 2007 Q1 through 2015 Q4. I kept domestic nonstop trips on a single coupon, dropped bulk fares, and kept trips with positive passenger counts, fares between \$50 and \$2,000, and distances between 50 and 5,000 miles. The two directions of an airport pair are combined into one route. This measures what people paid for tickets, not published fares, and not the full cost of travelling.

Route-average version. A route counts as an overlap if AirTran and Southwest each had at least 50 sampled passengers in 2009 Q2. Comparison routes had at least 50 AirTran passengers and no Southwest tickets at all. Requiring a fare in all 36 quarters leaves all 24 overlap routes and 106 possible comparison routes. Each overlap route is paired with one comparison route, never reused, matched on 2009 Q2 fare, passengers, distance, and how concentrated the route is among airlines. Pairs must agree on whether an endpoint is in Florida, and I cap how far apart a pair can be across those four traits combined at 2.5 (this is a limit on the combined gap, not on route mileage). 24 pairs survive. The usual balance check, the gap between the two groups divided by how spread out they are, comes to -0.31 on fare, +0.21 on distance and -0.24 on concentration. The other low-cost version is matched worse still, at +0.73 on distance, which is a reason to trust its bigger number less, not more.

Southwest's own fare version. The measure is what Southwest itself charged, so no other airline can affect it. Comparison routes are ones where Southwest and at least one airline other than AirTran both carried real traffic in 2009 Q2, and AirTran sold nothing. The second-airline requirement is necessary: Southwest's other routes are overwhelmingly Southwest-only, and without it the two groups cannot be matched on how concentrated they are. 137 routes qualify and 23 pairs match, with every balance check below 0.24. This match does not require pairs to agree on Florida. Requiring it leaves only 11 pairs and a -0.68 gap in Southwest's own starting fare. That is a real cost and I do not want to pretend otherwise: only 10 comparison routes touch Florida against 20 overlap routes, so this pool cannot give me both regional comparability and matched traits. Averaging four quarters handles season, not region. The version that forces regional matching is reported alongside rather than chosen.

Time windows, uncertainty, and the trend check

The starting point is **2009 Q3 to 2010 Q2**, the four quarters before the announcement. Every yearly result compares that year's four quarters against it, so all four seasons appear on both sides. For uncertainty I resample whole route pairs 10,000 times and report the middle 95% of the results. That captures ordinary sampling noise given the pairs I chose; it does not capture the risk that I chose the wrong pairs. The trend check fits a straight line through each pair's fare gap from 2007 Q1 to 2010 Q2, allowing for season.

Version and year	Pairs	Fare difference (95% range)
Route average vs. AirTran-only: 2013	24	+9.0% (+4.0%, +13.9%)
Route average vs. AirTran-only: 2015	24	+1.3% (-4.9%, +8.1%)
Route average vs. other low-cost: 2015	24	+14.8% (+7.9%, +22.1%)
Route average, matched on early trend: 2015	23	-0.4% (-6.6%, +6.5%)
Route average, connecting trips included: 2015	24	+1.8% (-4.3%, +8.3%)
Southwest's own fare: 2013	23	+14.0% (+8.6%, +20.1%)
Southwest's own fare: 2015	23	+12.0% (+6.7%, +17.9%)
Southwest's own fare, measured from 2008 Q3-2009 Q2 instead	23	+6.2% (+0.6%, +11.9%)

The last row matters. Southwest's fare gap was already +5.5% above the starting point back in 2008 Q3 to 2009 Q2, so measuring from that earlier window roughly halves the answer. The fitted trend line before the merger is flat, but the path itself is not, so the choice of starting point does real work. The row matched on early trend uses the fare path itself to pick comparison routes, which drags the later result toward the middle on its own; it is a sensitivity check, not a better estimate.

What each side would argue, and what neither can show

What someone arguing fares rose would point to

On 24 routes, two nonstop competitors became one. Once the mix problem is taken out, Southwest's own fares on those routes sit +12.0% above comparable Southwest routes in 2015 and +14.0% in 2013. The ranges stay above zero, the trend before the merger is flat, and the answer barely moves when I change the overlap definition or drop any single route. A GAO review describes a separate study of this merger by Le, which found 3.7% (range 2.1% to 5.3%) across a wider set of nonstop and connecting routes. That study used a different period, measure, and comparison group, and my figure is larger. [6,7]

What someone arguing they did not would point to

The DOJ expected new service and found no gate or slot problems at the overlap airports. [5] Southwest's own fare has a mix problem of its own: once it absorbed AirTran's passengers, the tickets inside its average were a different blend of fare types, booking times, and business travellers, and that alone can lift an average without any individual fare rising. Measuring from 2008 Q3 to 2009 Q2 instead of the four quarters before the announcement halves the answer to +6.2%. Every version that keeps the regions comparable leans on 8 to 11 Florida comparison routes used more than once, with starting fares well out of line, so the size rests on a handful of routes picked for one trait. The comparison routes were affected by the merger too, I do not measure seats or aircraft moves, and the route-average evidence is perfectly consistent with no lasting increase.

Why none of this proves the merger caused anything

- **The routes were not picked at random.** AirTran chose where to compete with Southwest, so overlap routes differ from other Southwest routes in ways my four matching traits do not catch. Geography is the clearest case: the overlap group is mostly a Florida network and the available comparison routes are mostly not.
- **The comparison routes were affected too.** Both groups changed under the merger, so what I measure is the extra difference on overlap routes, not a clean picture of a world without it.
- **Ticket mix still sits inside Southwest's own fare.** That measure is an average over Southwest's tickets, and which tickets those are changed when AirTran's passengers moved over.
- **The sample and the missing outcomes.** Requiring a fare in all 36 quarters keeps only routes that survived the whole period. Seats flown, fuel and airport costs, demand, bag and seat fees, connecting options, service quality, and any benefit on new routes are all outside this test.

Checks I ran

Including connecting trips as well as nonstop moves the route-average 2015 figure from +1.3% to +1.8%, and its trend before the merger is still +3.4% a year. Dropping one overlap route at a time leaves the route average between -0.3% and +2.8%, and Southwest's own fare between +10.8% and +13.0%. Requiring the smaller airline to hold 20% of route traffic moves the route average to +9.6% and Southwest's own fare to +12.0%. Southwest's own fare survives these checks. The route average does not.

What I would do next

None of these numbers should be turned into a damages figure. I would lead with the Southwest's-own-fare version and use the route average as an example of how the wrong measure can manufacture a result, then ask for the evidence that would close the gaps that remain: enough comparison routes to match Florida against Florida, ticket-level fare and booking data to separate price from who is flying, seat and aircraft records, and any internal pricing documents showing whether Southwest treated the old overlap routes differently. The most useful thing I found was that the appealing number was measuring the wrong thing.

Sources and notes

[1] BTS, Origin and Destination Survey (DB1B), a 10% quarterly sample of airline tickets. The files used here are the quarterly DB1BMarket archives named `Origin_and_Destination_Survey_DB1BMarket_YEAR_QUARTER.zip`, downloaded from transtats.bts.gov/PREZIP/. [2] Southwest, announcement, Sept. 27, 2010. [3] Southwest, close, May 2, 2011. [4] Southwest, final AirTran flight, departed Atlanta Dec. 28, 2014 (release dated Dec. 29). [5] DOJ, closing statement, Apr. 26, 2011. [6] Le, *An Empirical Analysis of the Price and Output Effects of the Southwest/AirTran Merger*, Competition and Regulation in Network Industries 17(3-4): 226-240 (2016), doi:10.1177/178359171601700302. [7] GAO, Airline Competition, GAO-26-107740 (2026).

The code, all 36 quarters of processed data, the matched route lists, and checksums for the original downloads come with this memo. Every number above is read from the results files when the document is built, so the text cannot drift away from the analysis.